

Denizeren Uzun

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Address: Bandırma, Türkiye (Home)

ABOUT ME

I'm a self-driven and result-oriented mechanical engineer with solid experience in the wind energy industry. Specialized in aerodynamic load calculation, project management, and data analysis of wind turbines, while bringing foundational knowledge in siting and turbine controller systems. Skilled in collaborating with cross-functional teams and project management to deliver high-quality results, whether refining designs or optimizing project outputs. Passionate about developing tools to enhance efficiency and streamline processes, with a proven track record of creating MATLAB and Excel based solutions that significantly reduce analysis time and increase quality. Currently pursuing a MSc in Energy Engineering with a thesis on degradation and predictive maintenance of onshore wind turbines, where I apply SCADA and condition monitoring data with machine learning to anticipate performance losses and availability risks. This work has strengthened my expertise in operational performance analysis, risk mitigation, and data-driven asset management.

WORK EXPERIENCE

SIEMENS GAMESA RENEWABLE ENERGY – İZMIR (HYBRID), TÜRKIYE

LOADS ENGINEER – 07/03/2021 – CURRENT

- **Wind Load Analysis & Optimization:** Analyzed wind conditions and load impacts to identify optimization potentials for large wind farms. Studied curtailments, deratings, wake meandering, power curve improvements and power boosts to optimize wind farms.
- **Site Specific Tower Design:** Load calculations for site-specific projects, contributing to potential savings with new and optimized towers. Acted as the focal point for site-specific tower loads, iterating designs based on load analysis to enhance structural integrity and reduce material use. In just one project saved costs up to 600,000€.
- **Tool Development & Efficiency:** Developed MATLAB and Excel-based tools that save 30+ minutes per load loop, enhance accuracy, and ensure error safety, minimizing manual mistakes.
- **Data Analysis & Machine Learning:** Worked on load emulators and data analysis tools to interpret load results and comparisons with component load envelopes.
- **Cross-Functional Collaboration:** Proactively communicated with design, siting, sales and external partners to gather feedback, strengthen relationships, and improve deliverables.
- **Engineering Quality & Project Management:** Participated in engineering quality reviews and received project management training to drive continuous improvement across workflows.

References:
Wouter van der Linde (Functional Manager): wouter.vanderlinde@siemensgamesa.com
Laura De La Fuente Gil (Former Functional Manager) laura.delafuente@siemensgamesa.com
Fırat Gür (Team Leader) gur.firat@yahoo.com
Bekir Han Biçim (Team Leader) bekir.bicim@siemensgamesa.com

ARÇELIK GLOBAL (COOKING APPLIANCES PLANT) – BOLU, TÜRKIYE

R&D DESIGN ENGINEER – 07/07/2019 – 28/02/2021

- **Product Owner:** Product owner of an oven developed for Scandinavian market, managing end-to-end development from design to manufacturing while balancing technical feasibility, cost, and reliability requirements.
- **Project Management:** Managed workload of a 4-person team (BOM Preparation), improving efficiency through prioritization and resource allocation.

- **R&D Focal Point:** Served as focal point between R&D and manufacturing, ensuring clear communication, quality assurance, and operational alignment.
- **Six Sigma:** Participated in a **Six Sigma improvement team**, gaining insights into process optimization and quality control.

EDUCATION AND TRAINING

19/09/2013 – 09/07/2020
MECHANICAL ENGINEERING Middle East Technical University

Taken Technical Electives:
- Design of Renewable Energy Systems
- Acoustics and Noise Control Engineering
- Dynamics of Machinery
- Internal Combustion Engine Design
- Engineering Economics

Field of study Engineering, Manufacturing, Economics | **Final grade** 2.55

03/2023 – CURRENT İzmir, Türkiye
ENERGY ENGINEERING MSC. İzmir Technology Institute

- Developing machine learning models using **SCADA and condition monitoring system (CMS) data** to predict production losses and availability drops in wind turbines.
- Studying **Conditional Health Monitoring** to apply into the maintenance schedules, and operations.
- Identified recurring fault patterns and performance degradation trends, **supporting proactive maintenance**.
- Applied data-driven asset management principles to quantify operational risks, optimize maintenance strategies, and improve long-term reliability of turbine components.
- Strengthened expertise in **data analytics, predictive modeling, and performance optimization**, directly transferable to asset management and operational performance monitoring roles.

Field of study Engineering, manufacturing and construction | **Final grade** 3.23 |

Thesis Degradation of Wind Turbines and Predictive Maintenance Applications with SCADA Data

30/01/2023 – 03/02/2023 Madrid, Spain
PROJECT MANAGEMENT Siemens Gamesa Renewable Energy A.Ş.

24/01/2021 – 04/02/2021
SIX SIGMA GREEN BELT Arçelik Global

12/02/2021 – 17/03/2021
ONLINE WIND ENERGY COURSE DTU-Technical university of Denmark

Website https://www.coursera.org/account/accomplishments/certificate/RY65RFAHJSLK

01/01/2022 – 08/02/2022
PYTHON ESSENTIALS Coursera

30/05/2021 – 24/06/2021
FUNDAMENTALS OF WIND ENERGY & WIND TURBINE TECHNOLOGY Middle East Technical University

Website https://sem.metu.edu.tr/upload/belge/2021/ODT.UE.K2100029.258-ruzgar-enerjisi-ve-ruzgar-turbini-teknolojileri-temel-online-denizeren-uzun.pdf

LANGUAGE SKILLS

Mother tongue(s): **TURKISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
GERMAN	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

MATLAB | ANSYS Mechanical | Siemens NX | Minitab | Microsoft Office | Python | Microsoft Excel VBA | QGIS: basic level | Prompt Engineering | BHawC | windPRO | project management | AGILE methodology | Backlog management/prioritization | machine learning | deep learning